Examiner only

7. (a) Explain what is meant by the term ‘multiwavelength astronomy’. [2]

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(b) The Stefan constant, σ, has the unit $W\,m^{-2}\,K^{-4}$. Express this in terms of base SI units. [2]

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(c) The wavelength of the peak emission of the bright star Arcturus is measured to be 674 nm.

(i) Calculate the star’s temperature. [2]

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(ii) The visible spectrum extends from approximately 400 nm to 700 nm. State the colour of Arcturus. [1]

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Examiner only

(d) A website claims that Arcturus is about 37 light years from the Sun. Use the following information and your answer to (c)(i) to justify this claim. [5]

Diameter of Arcturus = 2.78×10^{10} m
Intensity of radiation received on Earth from Arcturus = 3.09×10^{-8} W m^{-2}
1 light year = 9.46×10^{15} m

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END OF PAPER